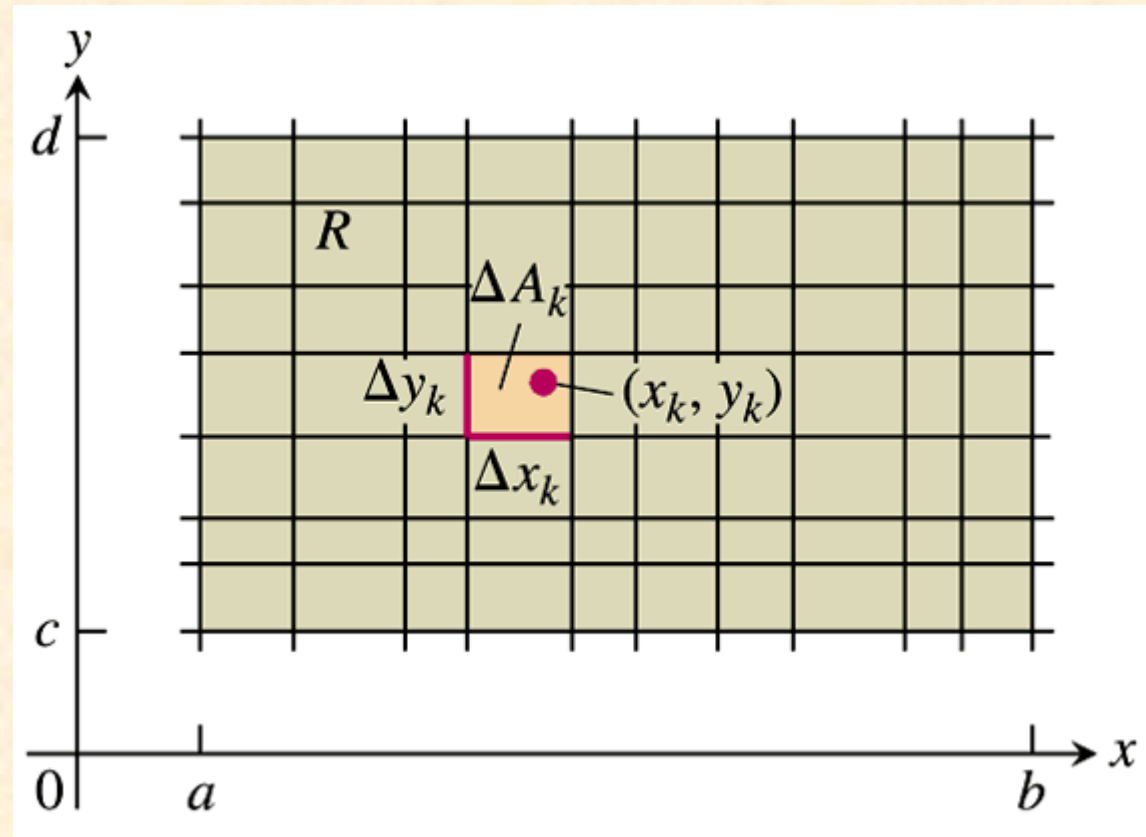


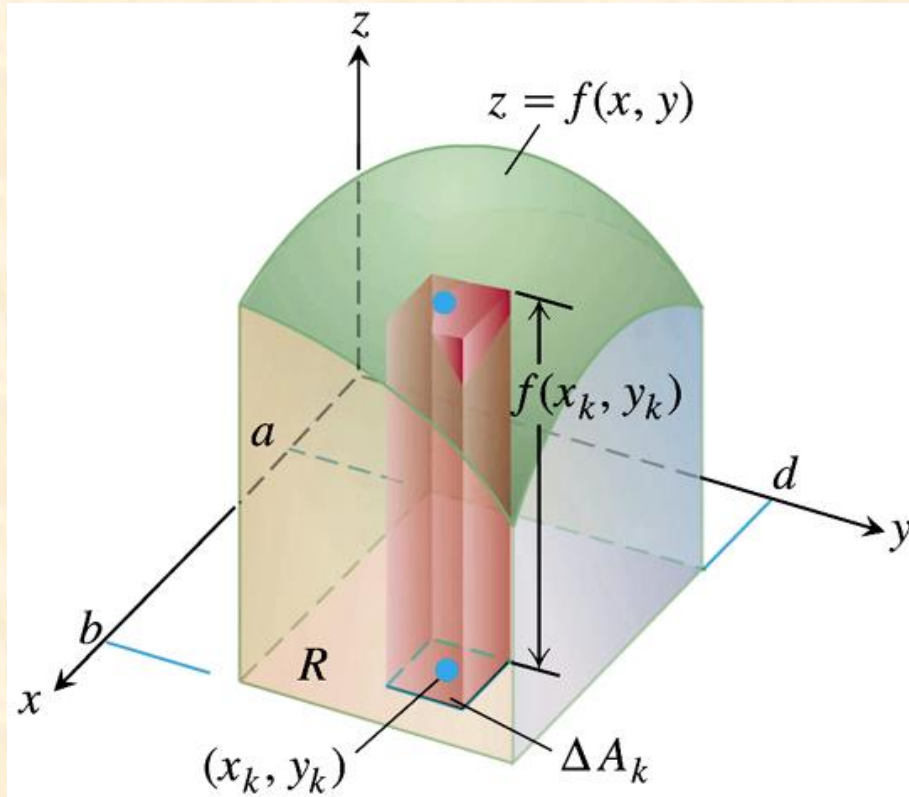
Double Integrals



$$S_n = \sum_{k=1}^n f(x_k, y_k) \cdot \Delta A_k$$

Rectangular grid partitioning the region R
into small rectangles of area $\Delta A_k = \Delta x_k \Delta y_k$

Double Integrals



$$\text{Volume} = S_n = \sum_{k=1}^n f(x_k, y_k) \cdot \Delta A_k$$

$$= \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k, y_k) \cdot \Delta A_k$$

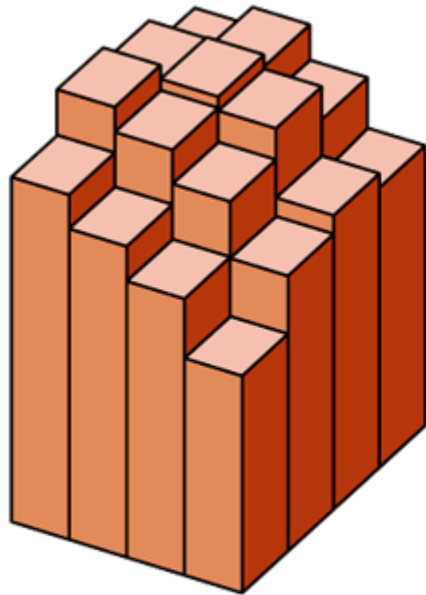
when $\Delta A_k \rightarrow 0$ and $n \rightarrow \infty$

$$= \iint_R f(x, y) \cdot dA$$

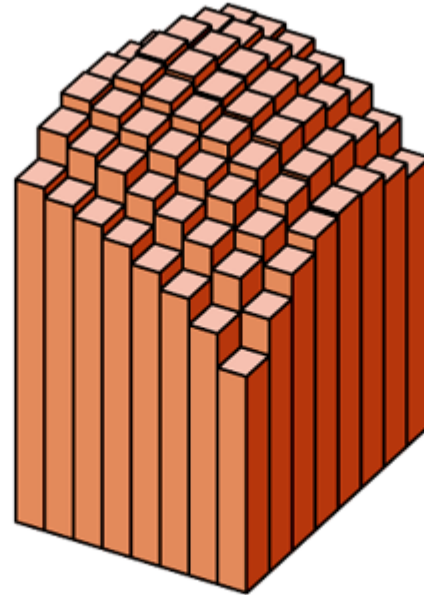
$$= \iint_R f(x, y) \cdot dx dy$$

Approximating solids with rectangular boxes leads us to define the volumes of more general solids as **double integrals**. The volume of the solid shown here is the double integral of $f(x, y)$ over the base region R .

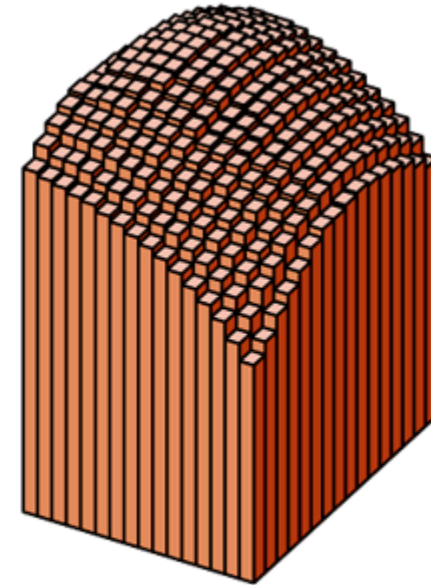
Double Integrals



(a) $n = 16$



(b) $n = 64$



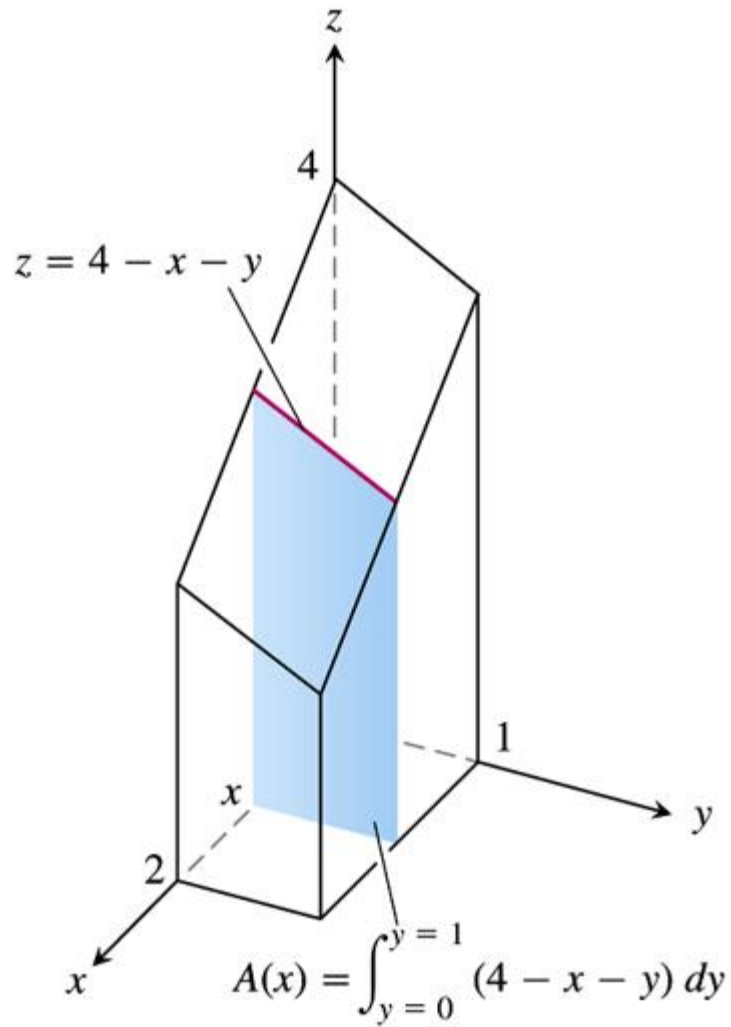
(c) $n = 256$

As n increases, the Riemann sum approximations approach the total volume of the solid.

Double Integrals

To obtain the cross-sectional area $A(x)$, we hold x fixed and integrate with respect to y .

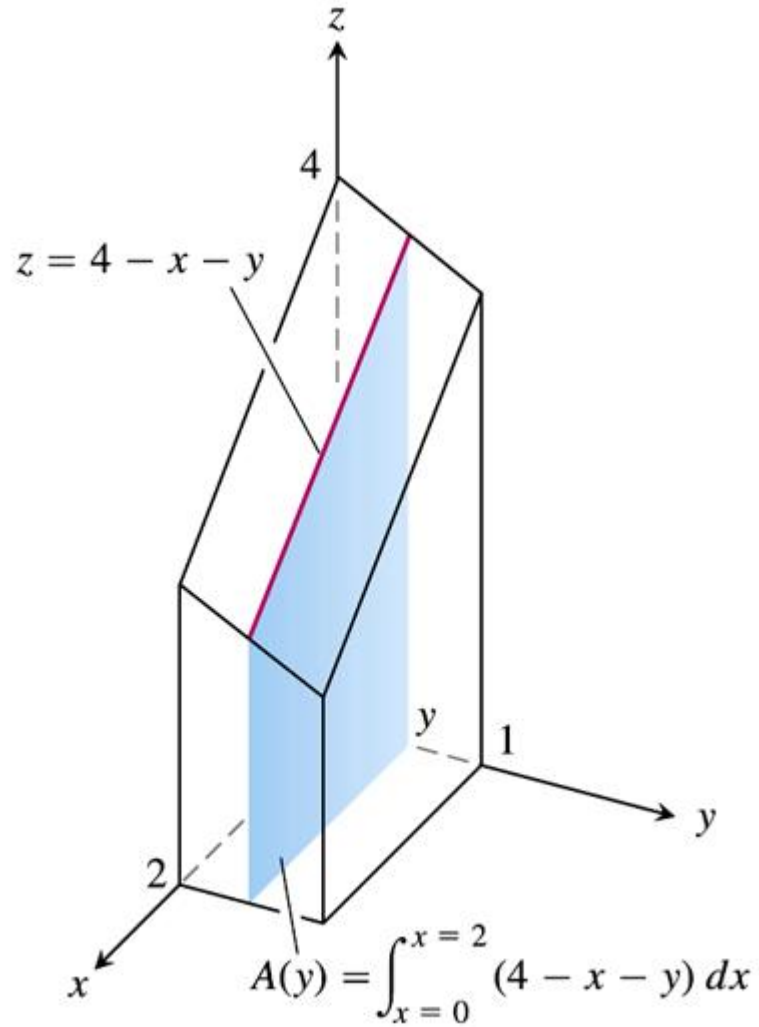
$$\begin{aligned}\text{Volume} &= \int_{x=0}^{x=2} A(x) dx \\ &= \int_{x=0}^{x=2} \left(\int_{y=0}^{y=1} (4 - x - y) dy \right) dx \\ &= \int_0^2 \int_0^1 (4 - y - x) dy dx \\ &= \dots\end{aligned}$$



Double Integrals

To obtain the cross-sectional area $A(x)$, we hold y fixed and integrate with respect *to* x .

$$\begin{aligned}\text{Volume} &= \int_{y=0}^{y=1} A(y) dy \\ &= \int_{y=0}^{y=1} \left(\int_{x=0}^{x=2} (4 - x - y) dx \right) dy \\ &= \int_0^2 \int_0^1 (4 - y - x) dx dy \\ &= \dots\end{aligned}$$



Double Integrals

THEOREM 1 Fubini's Theorem (First Form)

If $f(x, y)$ is continuous throughout the rectangular region $R: a \leq x \leq b, c \leq y \leq d$, then

$$\iint_R f(x, y) \, dA = \int_c^d \int_a^b f(x, y) \, dx \, dy = \int_a^b \int_c^d f(x, y) \, dy \, dx.$$

Double Integrals

THEOREM 14.1 Properties of Double Integrals

Let f and g be continuous over a closed, bounded plane region R , and let c be a constant.

1. $\iint_R cf(x, y) dA = c \iint_R f(x, y) dA$
2. $\iint_R [f(x, y) \pm g(x, y)] dA = \iint_R f(x, y) dA \pm \iint_R g(x, y) dA$
3. $\iint_R f(x, y) dA \geq 0$, if $f(x, y) \geq 0$
4. $\iint_R f(x, y) dA \geq \iint_R g(x, y) dA$, if $f(x, y) \geq g(x, y)$
5. $\iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA$, where R is the union of two nonoverlapping subregions R_1 and R_2 .

Double Integrals

EXAMPLE 1 Evaluating a Double Integral

Calculate $\iint_R f(x, y) \, dA$ for

$$f(x, y) = 1 - 6x^2y \quad \text{and} \quad R: \quad 0 \leq x \leq 2, \quad -1 \leq y \leq 1.$$

Solution By Fubini's Theorem,

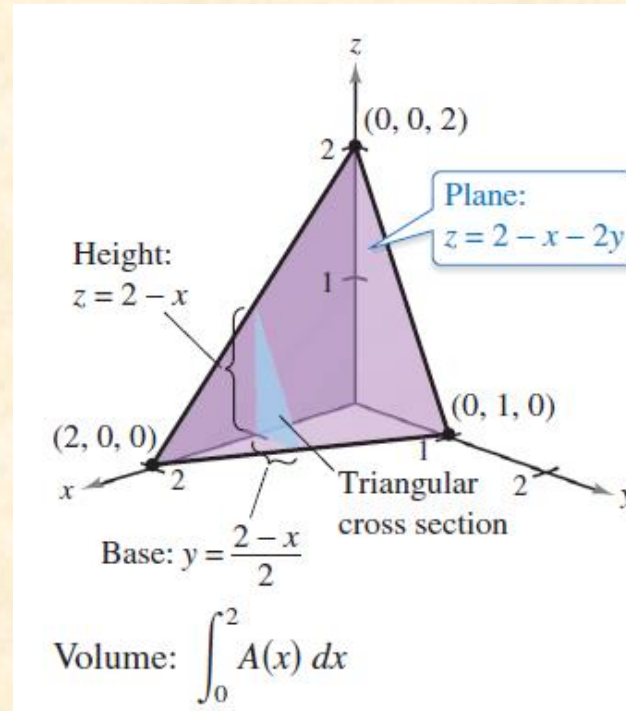
$$\begin{aligned} \iint_R f(x, y) \, dA &= \int_{-1}^1 \int_0^2 (1 - 6x^2y) \, dx \, dy = \int_{-1}^1 \left[x - 2x^3y \right]_{x=0}^{x=2} dy \\ &= \int_{-1}^1 (2 - 16y) \, dy = \left[2y - 8y^2 \right]_{-1}^1 = 4. \end{aligned}$$

Reversing the order of integration gives the same answer:

$$\begin{aligned} \int_0^2 \int_{-1}^1 (1 - 6x^2y) \, dy \, dx &= \int_0^2 \left[y - 3x^2y^2 \right]_{y=-1}^{y=1} dx \\ &= \int_0^2 [(1 - 3x^2) - (-1 - 3x^2)] \, dx \\ &= \int_0^2 2 \, dx = 4. \end{aligned}$$

Double Integrals

Example 2: Consider the solid region bounded by the plane $z = f(x, y) = 2 - x - 2y$ and the three coordinate planes, as shown in Figure 14.14.

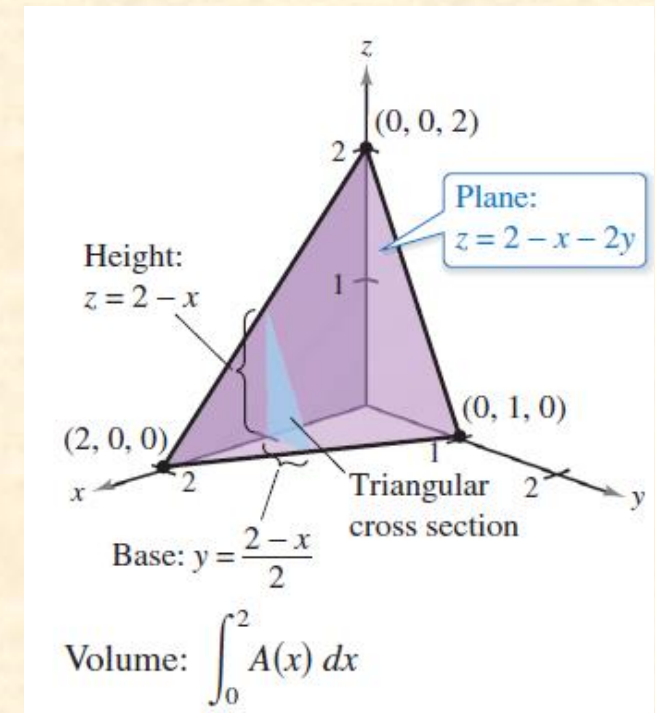


Double Integrals

Each vertical cross section taken parallel to the yz -plane is a triangular region whose base has a length of $y = (2 - x)/2$ and whose height is $z = 2 - x$.

This implies that for a fixed value of x , the area of the triangular cross section is

$$A(x) = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}\left(\frac{2 - x}{2}\right)(2 - x) = \frac{(2 - x)^2}{4}.$$



Double Integrals

By the formula for the volume of a solid with **known cross sections**, the volume of the solid is

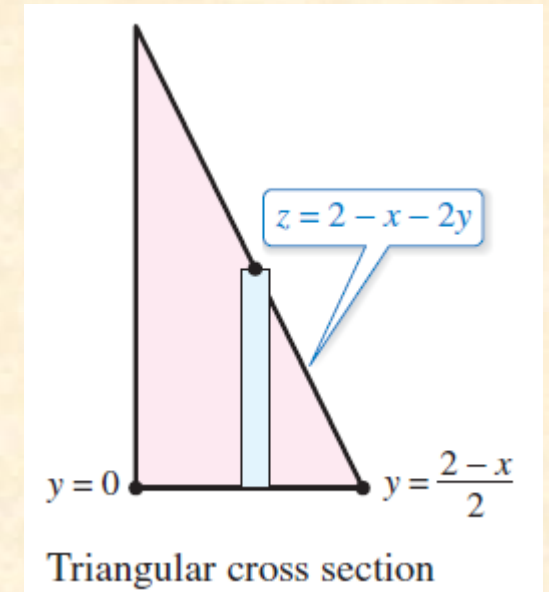
$$\begin{aligned}\text{Volume} &= \int_a^b A(x) \, dx \\ &= \int_0^2 \frac{(2-x)^2}{4} \, dx \\ &= \left. -\frac{(2-x)^3}{12} \right|_0^2 \\ &= \frac{2}{3}.\end{aligned}$$

Formula for volume

Substitute.

Integrate with respect to x .

Volume of solid region



This procedure works no matter how $A(x)$ is obtained. In particular, you can find $A(x)$ by integration, as shown in Figure 14.15.

Double Integrals

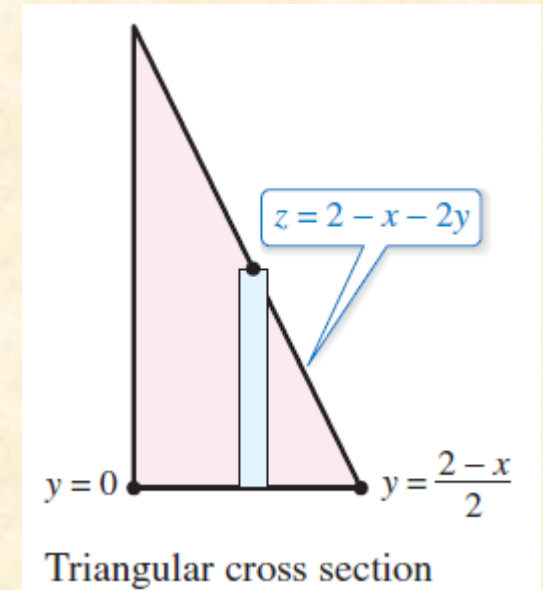
That is, you consider x to be constant and integrate $z = 2 - x - 2y$ from 0 to $(2 - x)/2$ to obtain

$$\begin{aligned} A(x) &= \int_0^{(2-x)/2} (2 - x - 2y) dy \\ &= \left[(2 - x)y - y^2 \right]_0^{(2-x)/2} \\ &= \frac{(2 - x)^2}{4}. \end{aligned}$$

Apply formula for area.

Integrate with respect to y .

Area of triangular cross section



Combining these results, you have the *iterated integral*

$$\text{Volume} = \iint_R f(x, y) dA = \int_0^2 \int_0^{(2-x)/2} (2 - x - 2y) dy dx.$$

Double Integrals

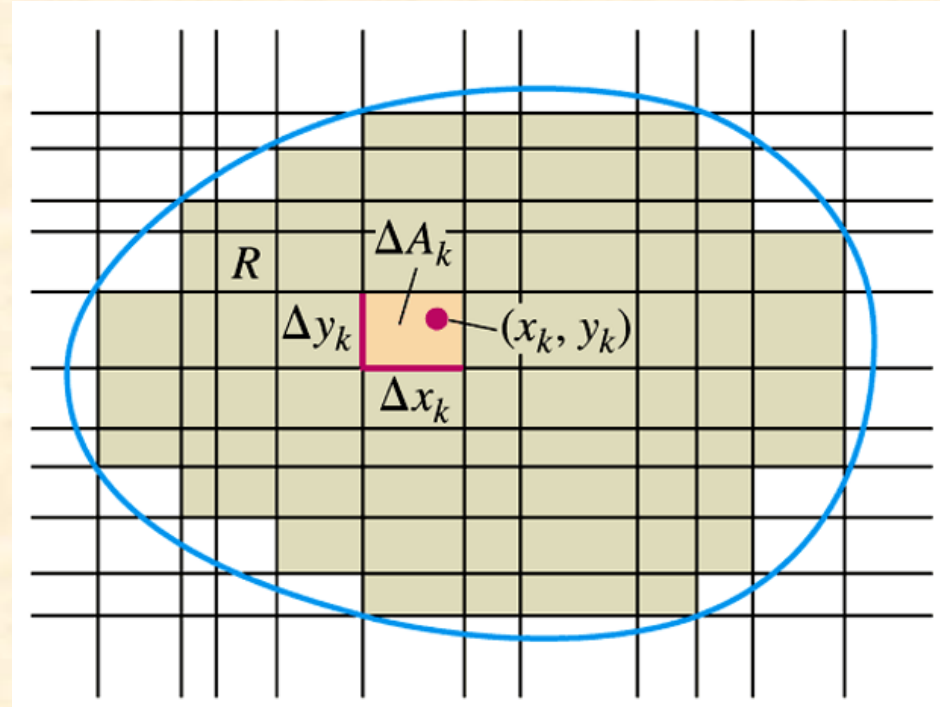
Practice Problem:

1. Find the volume of the region bounded above by the elliptical paraboloid $z = 10 + x^2 + 3y^2$ and below by the rectangle $R: 0 \leq x \leq 1, 0 \leq y \leq 2$.

2. Evaluate $\iint_R \left(1 - \frac{1}{2}x^2 - \frac{1}{2}y^2\right) dA$

where R is the region given by $0 \leq x \leq 1, 0 \leq y \leq 1$.

Double Integrals



A rectangular grid partitioning a bounded nonrectangular region into rectangular cells.

Double Integrals

THEOREM 2 Fubini's Theorem (Stronger Form)

Let $f(x, y)$ be continuous on a region R .

1. If R is defined by $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$, with g_1 and g_2 continuous on $[a, b]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

2. If R is defined by $c \leq y \leq d$, $h_1(y) \leq x \leq h_2(y)$, with h_1 and h_2 continuous on $[c, d]$, then

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

Double Integrals

Example 3: Find the volume of the prism whose base is the triangle in the xy -plane bounded by the x -axis and the lines $y = x$ and $x = 1$ and whose top lies in the plane $z = f(x, y) = 3 - x - y$.

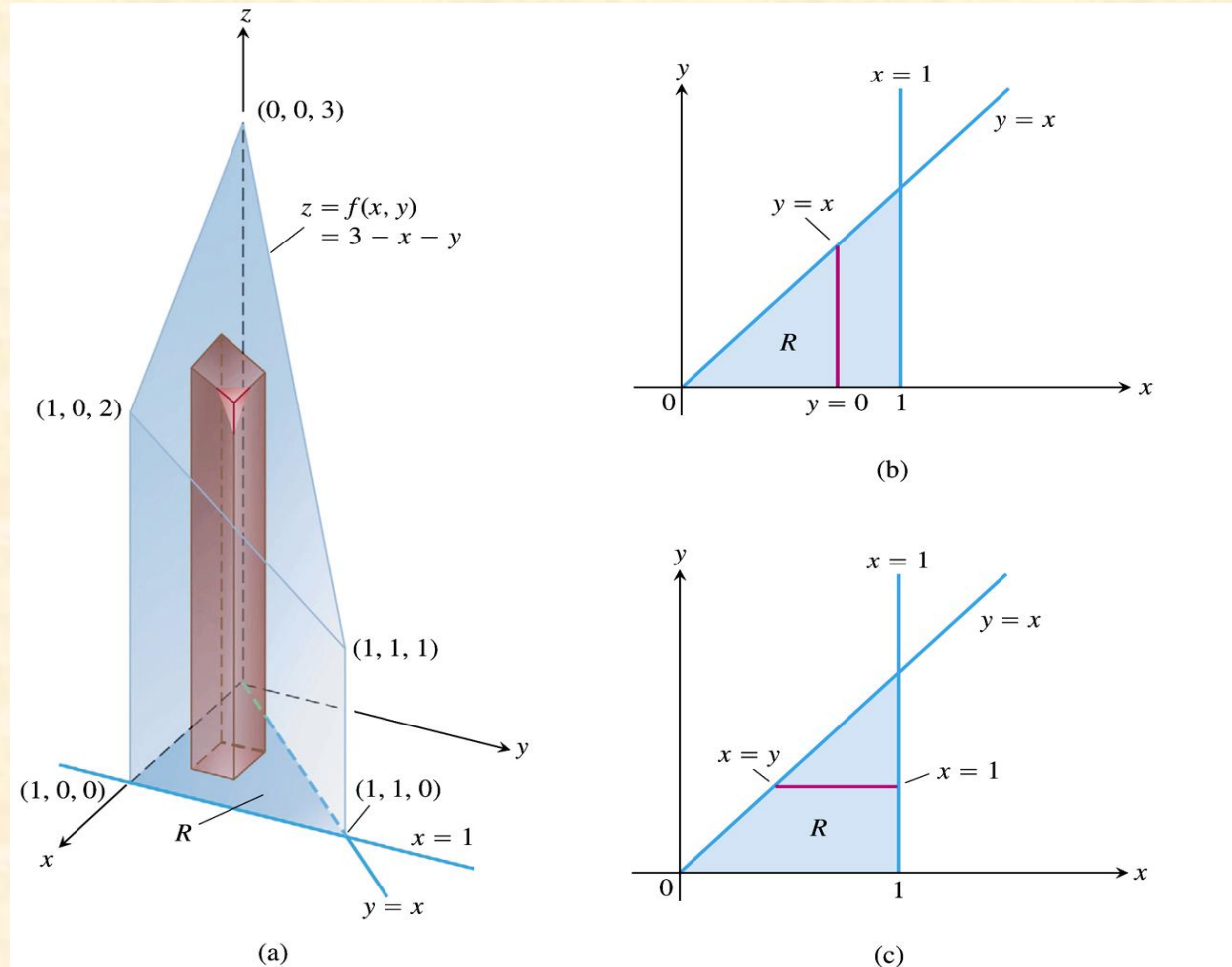


FIGURE 15.11 (a) Prism with a triangular base in the xy -plane. The volume of this prism is defined as a double integral over R . To evaluate it as an iterated integral, we may integrate first with respect to y and then with respect to x , or the other way around (Example 2).
(b) Integration limits of

Double Integrals

EXAMPLE 2 Finding Volume

Find the volume of the prism whose base is the triangle in the xy -plane bounded by the x -axis and the lines $y = x$ and $x = 1$ and whose top lies in the plane

$$z = f(x, y) = 3 - x - y.$$

Solution See Figure 15.11 on page 1075. For any x between 0 and 1, y may vary from $y = 0$ to $y = x$ (Figure 15.11b). Hence,

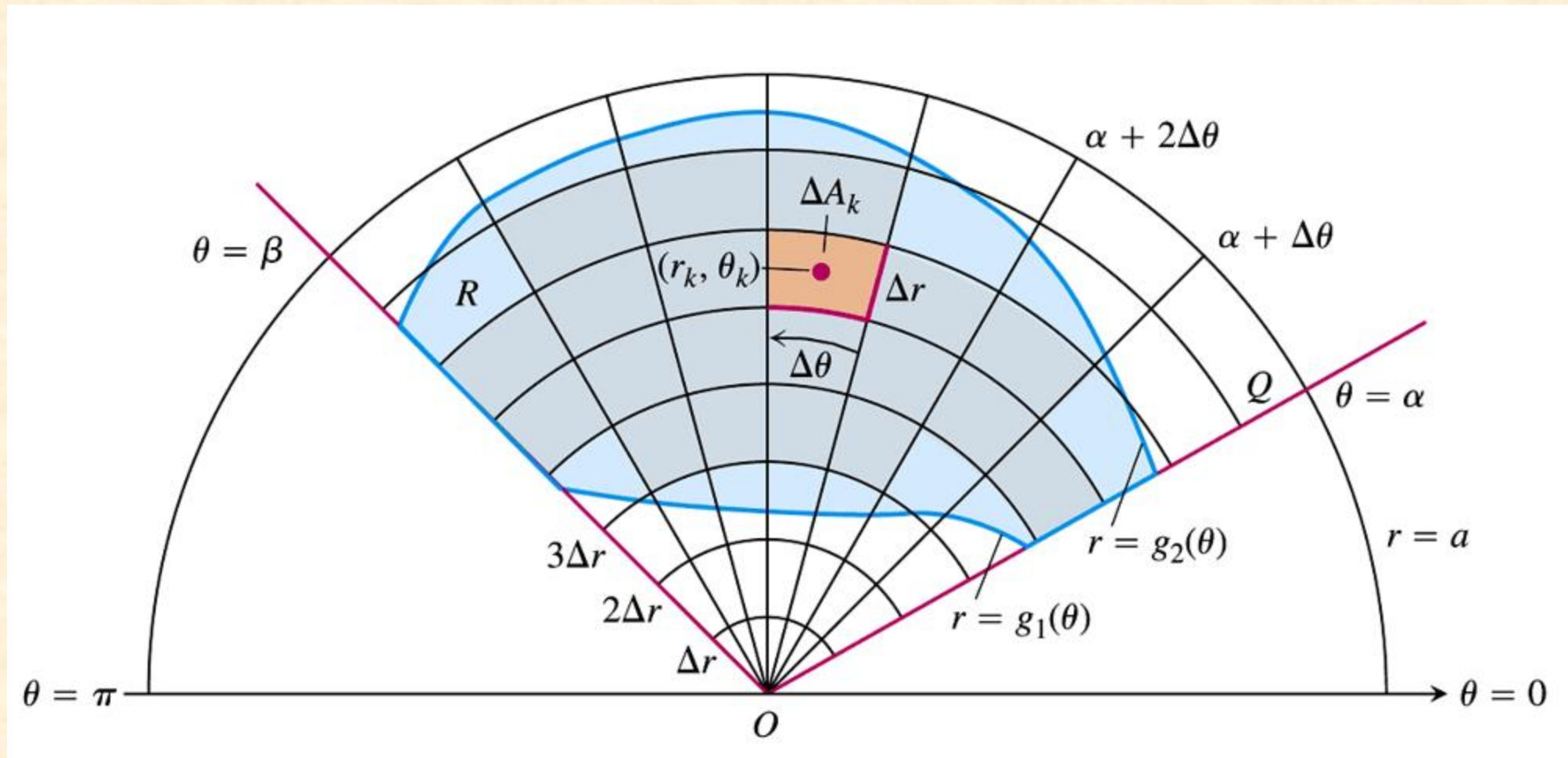
$$\begin{aligned} V &= \int_0^1 \int_0^x (3 - x - y) dy dx = \int_0^1 \left[3y - xy - \frac{y^2}{2} \right]_{y=0}^{y=x} dx \\ &= \int_0^1 \left(3x - \frac{3x^2}{2} \right) dx = \left[\frac{3x^2}{2} - \frac{x^3}{2} \right]_{x=0}^{x=1} = 1. \end{aligned}$$

When the order of integration is reversed (Figure 15.11c), the integral for the volume is

$$\begin{aligned} V &= \int_0^1 \int_y^1 (3 - x - y) dx dy = \int_0^1 \left[3x - \frac{x^2}{2} - xy \right]_{x=y}^{x=1} dy \\ &= \int_0^1 \left(3 - \frac{1}{2} - y - 3y + \frac{y^2}{2} + y^2 \right) dy \\ &= \int_0^1 \left(\frac{5}{2} - 4y + \frac{3}{2}y^2 \right) dy = \left[\frac{5}{2}y - 2y^2 + \frac{y^3}{2} \right]_{y=0}^{y=1} = 1. \end{aligned}$$

The two integrals are equal, as they should be.

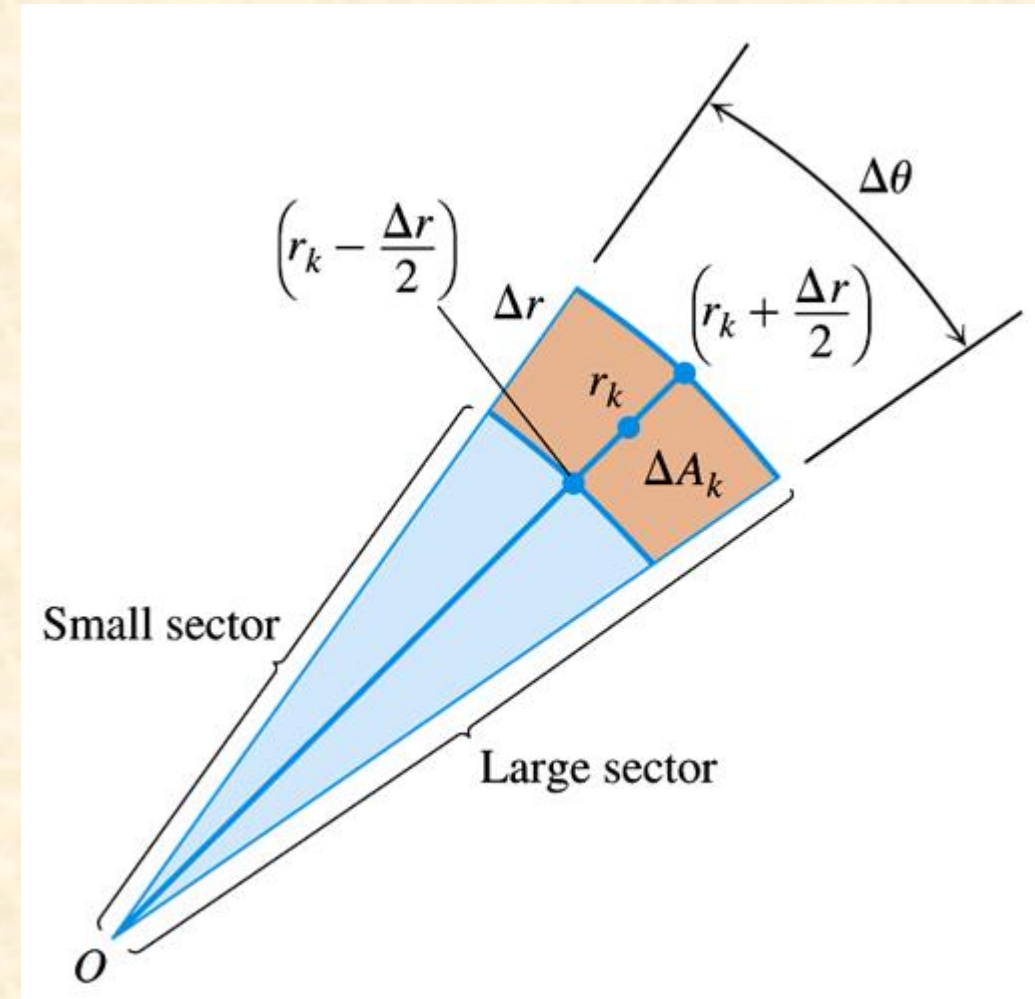
Double Integrals in Polar Form



Double Integrals in Polar Form

The area of a wedge-shaped sector of a circle having radius r and angle θ is $A = \frac{1}{2} \cdot \theta \cdot r^2$

Inner radius:	$\frac{1}{2} \left(r_k - \frac{\Delta r}{2} \right)^2 \Delta \theta$
Outer radius:	$\frac{1}{2} \left(r_k + \frac{\Delta r}{2} \right)^2 \Delta \theta.$



Double Integrals in Polar Form

Therefore,

$$\begin{aligned}\Delta A_k &= \text{area of large sector} - \text{area of small sector} \\ &= \frac{\Delta\theta}{2} \left[\left(r_k + \frac{\Delta r}{2} \right)^2 - \left(r_k - \frac{\Delta r}{2} \right)^2 \right] = \frac{\Delta\theta}{2} (2r_k \Delta r) = r_k \Delta r \Delta\theta.\end{aligned}$$

Combining this result with the sum defining S_n gives

$$S_n = \sum_{k=1}^n f(r_k, \theta_k) r_k \Delta r \Delta\theta.$$

As $n \rightarrow \infty$ and the values of Δr and $\Delta\theta$ approach zero, these sums converge to the double integral

$$\lim_{n \rightarrow \infty} S_n = \iint_R f(r, \theta) r \, dr \, d\theta.$$

Double Integrals in Polar Form

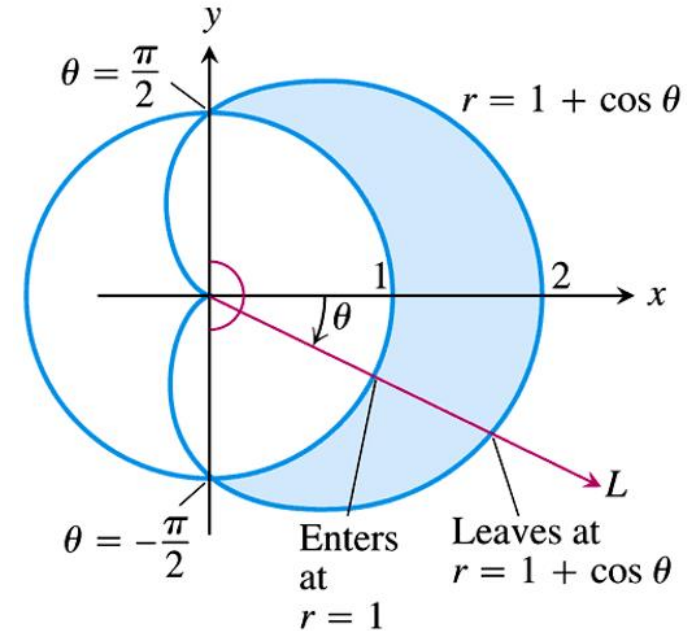
EXAMPLE 1 Finding Limits of Integration

Find the limits of integration for integrating $f(r, \theta)$ over the region R that lies inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 1$.

Solution

1. We first sketch the region and label the bounding curves (Figure 15.23).
2. Next we find the *r*-limits of integration. A typical ray from the origin enters R where $r = 1$ and leaves where $r = 1 + \cos \theta$.
3. Finally we find the *θ* -limits of integration. The rays from the origin that intersect R run from $\theta = -\pi/2$ to $\theta = \pi/2$. The integral is

$$\int_{-\pi/2}^{\pi/2} \int_1^{1+\cos \theta} f(r, \theta) r \, dr \, d\theta.$$



Double Integrals in Polar Form

Area in Polar Coordinates

The area of a closed and bounded region R in the polar coordinate plane is

$$A = \iint_R r \, dr \, d\theta.$$

THEOREM 14.3 Change of Variables to Polar Form

Let R be a plane region consisting of all points $(x, y) = (r \cos \theta, r \sin \theta)$ satisfying the conditions $0 \leq g_1(\theta) \leq r \leq g_2(\theta)$, $\alpha \leq \theta \leq \beta$, where $0 \leq (\beta - \alpha) \leq 2\pi$. If g_1 and g_2 are continuous on $[\alpha, \beta]$ and f is continuous on R , then

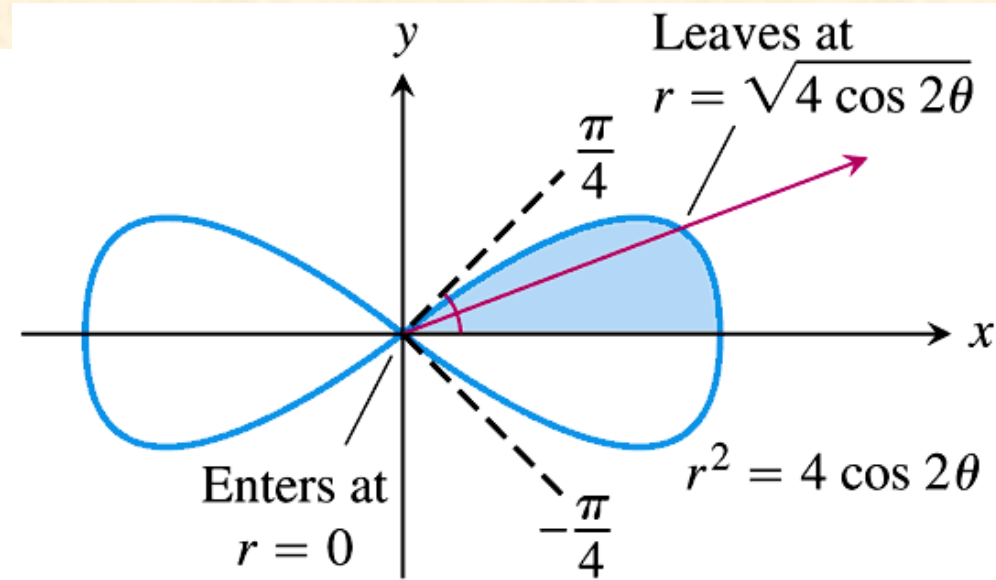
$$\iint_R f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta.$$

Double Integrals in Polar Form

EXAMPLE 2 Find the area enclosed by the lemniscate $r^2 = 4 \cos 2\theta$.

Solution We graph the lemniscate to determine the limits of integration (Figure 15.25) and see from the symmetry of the region that the total area is 4 times the first-quadrant portion.

$$\begin{aligned} A &= 4 \int_0^{\pi/4} \int_0^{\sqrt{4 \cos 2\theta}} r \, dr \, d\theta = 4 \int_0^{\pi/4} \left[\frac{r^2}{2} \right]_{r=0}^{r=\sqrt{4 \cos 2\theta}} d\theta \\ &= 4 \int_0^{\pi/4} 2 \cos 2\theta \, d\theta = 4 \sin 2\theta \Big|_0^{\pi/4} = 4. \end{aligned}$$



■ **FIGURE 15.24** To integrate over the shaded region, we run r from 0 to $\sqrt{4 \cos 2\theta}$ and θ from 0 to $\pi/4$ (Example 2).

Double Integrals in Polar Form

EXAMPLE 3 Evaluate

$$\iint_R e^{x^2+y^2} dy dx,$$

where R is the semicircular region bounded by the x -axis and the curve $y = \sqrt{1-x^2}$ (Figure 15.26).

Solution In Cartesian coordinates, the integral in question is a **nonelementary integral** and there is **no direct way to integrate** $e^{x^2+y^2}$ with respect to either x or y . Yet this integral and others like it are important in mathematics—in statistics, for example—and we need to find a way to evaluate it. Polar coordinates save the day. Substituting $x = r \cos \theta$, $y = r \sin \theta$ and replacing $dy dx$ by $r dr d\theta$ enables us to evaluate the integral as

$$\begin{aligned}\iint_R e^{x^2+y^2} dy dx &= \int_0^\pi \int_0^1 e^{r^2} r dr d\theta = \int_0^\pi \left[\frac{1}{2} e^{r^2} \right]_0^1 d\theta \\ &= \int_0^\pi \frac{1}{2} (e - 1) d\theta = \frac{\pi}{2} (e - 1).\end{aligned}$$

Double Integrals in Polar Form

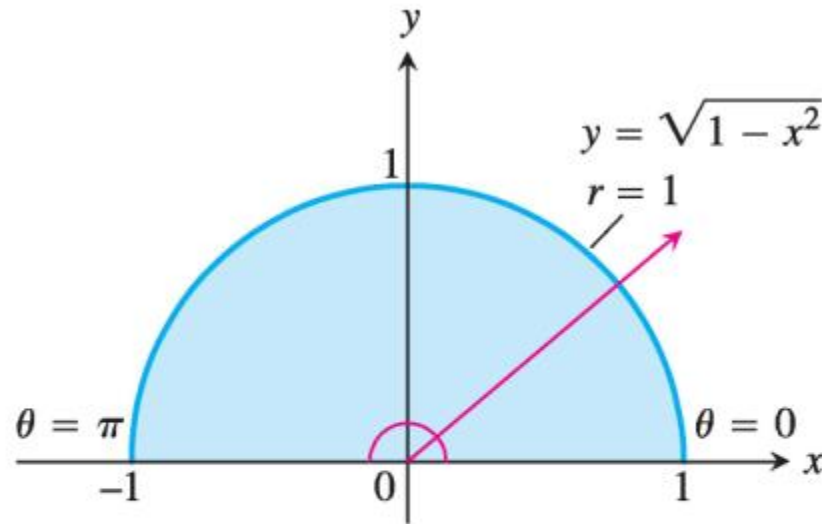


FIGURE 15.26 The semicircular region in Example 3 is the region

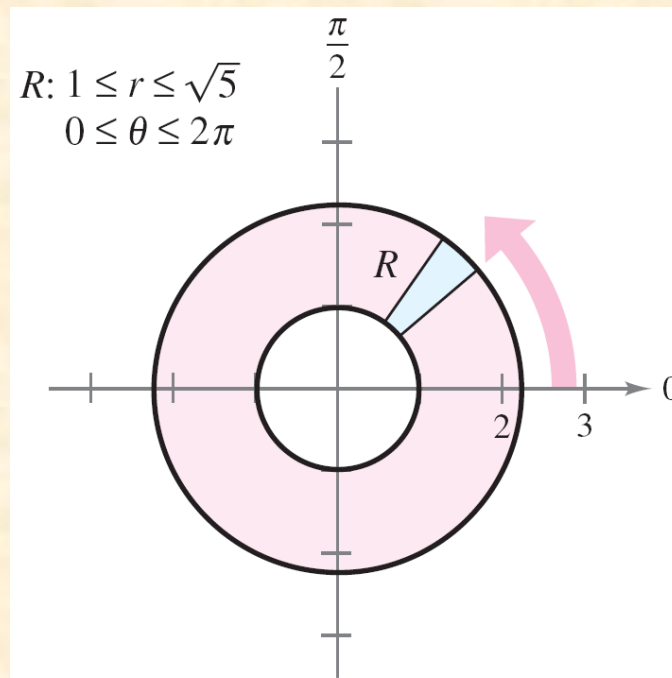
$$0 \leq r \leq 1, \quad 0 \leq \theta \leq \pi.$$

Double Integrals in Polar Form

Problem 4: Let R be the annular region lying between the $x^2 + y^2 = 1$ and $x^2 + y^2 = 5$. Evaluate the integral $\int_R (x^2 + y) dA$.

Solution:

The polar boundaries are $1 \leq r \leq \sqrt{5}$ and $0 \leq \theta \leq 2\pi$, as shown in Figure 14.29.



Double Integrals in Polar Form

Furthermore, $x^2 = (r \cos \theta)^2$ and $y = r \sin \theta$.

So, you have

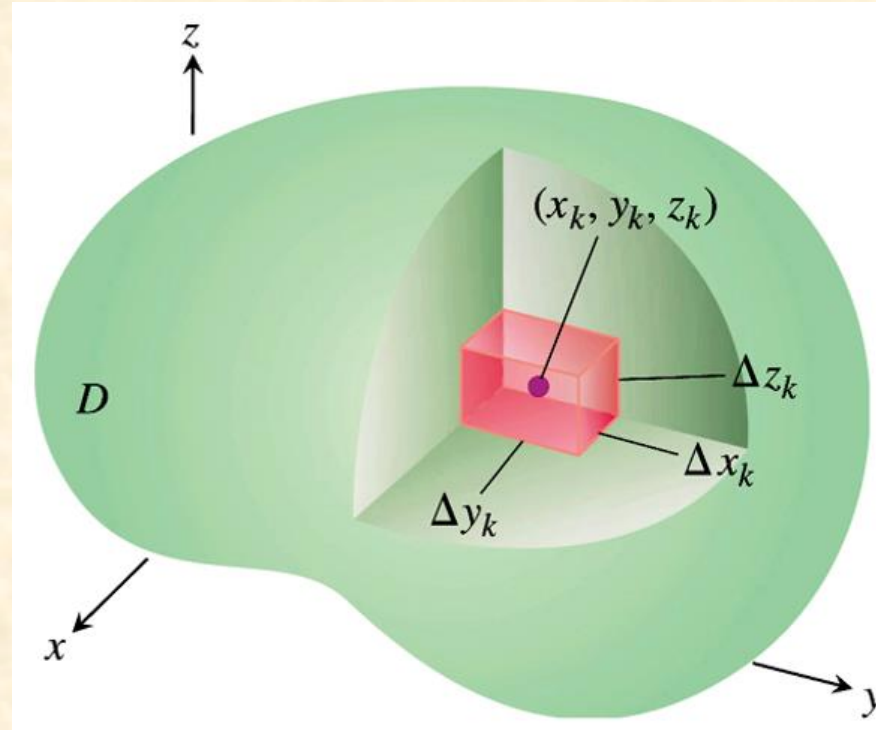
$$\iint_R (x^2 + y) dA = \int_0^{2\pi} \int_1^{\sqrt{5}} (r^2 \cos^2 \theta + r \sin \theta) r dr d\theta = \int_0^{2\pi} \int_1^{\sqrt{5}} (r^3 \cos^2 \theta + r^2 \sin \theta) dr d\theta$$

$$= \int_0^{2\pi} \left(\frac{r^4}{4} \cos^2 \theta + \frac{r^3}{3} \sin \theta \right) \Big|_1^{\sqrt{5}} d\theta = \int_0^{2\pi} \left(6 \cos^2 \theta + \frac{5\sqrt{5} - 1}{3} \sin \theta \right) d\theta$$

$$= \int_0^{2\pi} \left(3 + 3 \cos 2\theta + \frac{5\sqrt{5} - 1}{3} \sin \theta \right) d\theta = \left(3\theta + \frac{3 \sin 2\theta}{2} - \frac{5\sqrt{5} - 1}{3} \cos \theta \right) \Big|_0^{2\pi}$$

$$= 6\pi.$$

Triple Integrals in Rectangular Coordinates



DEFINITION Volume

The **volume** of a closed, bounded region D in space is

$$V = \iiint_D dV.$$

Triple Integrals in Rectangular Coordinates

EXAMPLE 1 Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Solution The volume is

$$V = \iiint_D dz \, dy \, dx,$$

Triple Integrals in Rectangular Coordinates

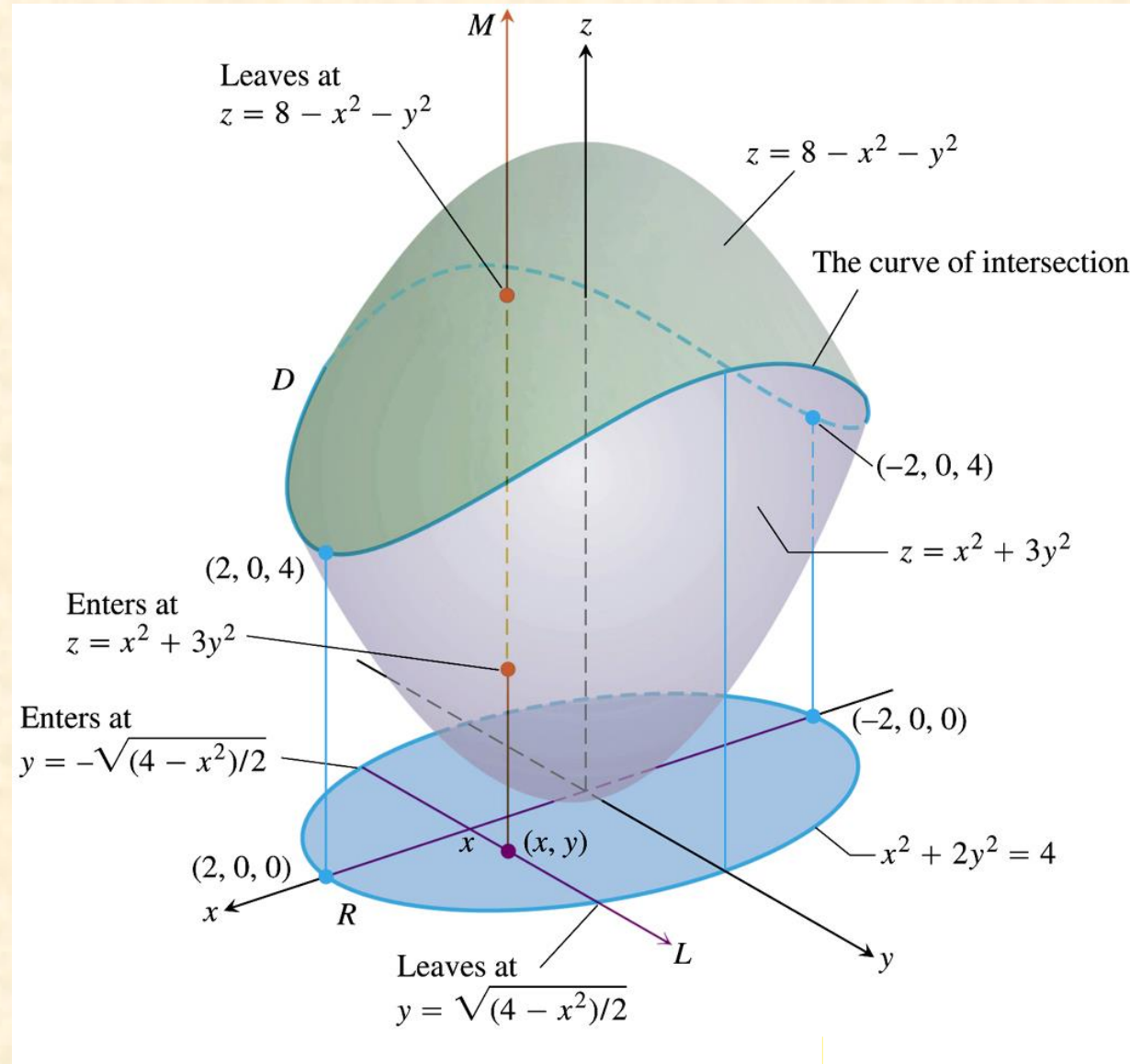


FIGURE 15.28 The volume of the region enclosed by two paraboloids, calculated in Example 1.

Thanks a lot ...